

Video Question Answering and Beyond

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Abstract

Video Question Answering (Video QA) has emerged as a central task in multimodal learning. This tutorial provides a comprehensive overview of VideoQA research and highlights new frontiers. We begin with an introduction to VideoQA preliminaries, tracing how methods have adapted from third-person view short videos to capture egocentric and long-ranged spatial-temporal dynamics. We then focus on the impact of large multimodal models. Next, we expand the scope to spatial understanding beyond videos. Each topic is discussed through the lens of tasks, datasets, methods, and evaluation protocols. Finally, we conclude with future directions, including fine-grained and long-ranged video understanding, robustness and trustworthiness, Egocentric and embodied assistance, and omnimodal integration. This tutorial aims to equip participants with both a historical perspective and a forward-looking roadmap for advancing Video QA in the LLM era.

CCS Concepts

• **Computing methodologies** → **Video summarization; Scene understanding.**

Keywords

Video Question Answering, Video-language Understanding, Multimodal LLMs

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1 Tutorial Outline

This tutorial will 1) introduce, review, and summarize recent advances in video question answering along with its related topics, and 2) provide insights into the key challenges and potential future research directions in this field. Specifically, it covers 4 key sessions.

1.1 Session I: Introduction and Overview

In this session, we will introduce the VideoQA task definition, its research and practical significance, as well as a brief overview of the VideoQA technique evolution.

1.2 Session II: VideoQA in LLM Era

In this session, we focus on LLM-driven VideoQA techniques and benchmarks [9, 10, 19]. The techniques cover both foundation multimodal LLMs [2] and Socratic approaches [23]. We will analyze different methods from the aspect of data, architectures, and training strategies if available.

The benchmarks include those emphasizing temporal understanding, long video understanding [8], streaming video understanding [7], and egocentric video understanding [15, 20]. We will analyze the unique challenges of different benchmarks, along with representative solutions.

1.3 Session III: Spatial Understanding in VideoQA

This session expands VideoQA into spatial reasoning beyond conventional 2D settings. We will cover **3D Visual Question Answering** [1, 14], which grounds language queries in reconstructed 3D environments, and **Multiview and Panoramic QA** [3, 16, 24], which require integrating information across multiple viewpoints or 360-degree scenes.

1.4 Session IV: Future Directions

The final session looks forward to open challenges and opportunities in VideoQA research. We will discuss **long and fine-grained video understanding** [12, 13, 18], **robustness and trustworthiness** in Video Understanding models models [11, 19, 21, 25], the role of **egocentric and embodied assistance** for agents [4–6, 17], and prospects for truly **omnimodal interaction** across vision, language, audio, and beyond [22].

2 Specification of the Tutorial

Target Audience: The tutorial is designed for researchers, graduate students, and practitioners in computer vision, natural language processing, and multimodal AI who are interested in VideoQA and its emerging extensions. We particularly expect 50 to 100 participants from the areas of video understanding, large multimodal models (LMMs), embodied AI, and robotics.

Prerequisites:

- Basic knowledge of machine learning and deep learning.
- Familiarity with transformers and recent advances in vision-language models.
- Prior exposure to video understanding or QA tasks will be helpful but not required.

Duration: 2.5 hours (approximately 150 minutes).

Material Resources:

- **Slides:** Comprehensive tutorial slides will be provided to participants in PDF format.
- **Survey Notes:** A curated reading list of representative papers, datasets, and benchmarks (with BibTeX).
- **Code Resources:** Links to open-source toolkits and repositories (e.g., HuggingFace, PyTorch-based implementations of VideoQA methods).
- **Demo Materials:** Short demos showing VideoQA pipelines, dataset examples, and benchmark evaluations.

3 Tutorial Presenters

Yicong Li is currently a Research Fellow at the National University of Singapore (NUS). His research interests lie in multimodal learning, 2D/3D generation, and trustworthy AI. He has published papers at top-tier conferences and journals including CVPR, ICCV, ICLR, and TPAMI. His recent works focus on 3D understanding and generative models.

Junbin Xiao is currently a Research Fellow at the National University of Singapore (NUS). His research focuses on video QA and trustworthy multimodality. He devotes himself to developing AI techniques that can understand the physical world, interact and communicate with human beings to provide personalized assistance. He has good publications on top-tier venues (e.g., TPAMI, IJCV, CVPR, ICCV, ECCV) and serves as reviewer for them.

Angela Yao is an Dean's Chair Associate Professor and leads the CVML group at the National University of Singapore. Her research spans computer vision, machine learning, and multimodal reasoning, with a particular focus on human motion analysis and video understanding. She has published extensively in computer vision (CVPR, ICCV, ECCV) and machine learning (NeurIPS, ICML, ICLR) conferences, and serves as general chair or as area chair for them.

Tat-Seng Chua is the KITHCT Chair Professor at National University of Singapore. He is elected as a Singapore National Academy of Science (SNAS) Fellow. He has works in a wide range of leading conferences and scholarly publications, with over 93K citations. His main research interests include unstructured data analytics, video analytics, conversational search and recommendation, and robust and trustable AI. Prof. Chua holds a PhD from the University of Leeds, UK.

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